

Two paralog pairs, two fates — set by reads, not % identity

Long HiFi reads multimap between paralog copies; what matters is whether a read covers a copy-distinguishing position (PSV).

RABL2 (best case)

RABL2A / RABL2B — 99.9% identical mRNA

197 / 200 reads multimap, every one DECISIVE

RABL2A  151 decisive

RABL2B  49 decisive

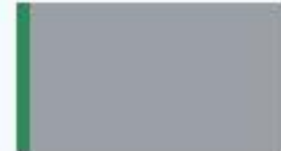
✓ **0 tied → BOTH copies resolvable**

DAZ (hardest case)

DAZ1 / DAZ3 — identical shared backbone

217 / 220 reads multimap; markers off the read

DAZ1  174 decisive

DAZ3  2 decisive +41 tied

✗ **DAZ3: 2 decisive, 41 tied → PHANTOM**

Green = reads that align strictly better to this copy (decisive own evidence). Grey = reads equally good on both copies (tied → undecidable).

Why it works: decisive reads = PSVs the read actually covers

RABL2 — differences fall in exons the read spans

one HiFi read



copy A (true origin)



copy B (paralog)



read covers 3 PSVs → matches A, mismatches B → **DECISIVE** for A ✓

DAZ — read lands on the identical shared backbone

one HiFi read



DAZ1



DAZ3



0 PSVs covered → identical score on both → **TIED (coin-flip)** ✗

rustle's decisive-evidence guard — 3 buckets, never fabricate

AUTHENTIC

≥k reads decisive
for this copy

→ *RABL2A, RABL2B*
(real recovery)

UNRESOLVABLE

only tied reads
(no decisive)

→ *tied DAZ reads*
(can't prove/deny)

PHANTOM

coverage is sister
spillover only

→ *DAZ3*
(refuse to emit)

StringTie is sequence-blind & drops secondaries → cannot tell these apart. The win is correct apportionment + refusing to fabricate, not raw copy count.

Predicting identifiability from minimap2's minimizers (reference-only)

minimap2 seeds on canonical minimizers (k19/w19). Shared minimizers → read multimaps. A DISCRIMINATIVE minimizer (in one copy only) is what lets it tell them apart.



RABL2A vs RABL2B — 99.9% identical

158/273 minimizers discriminative (57%) — densely spread, no read can avoid one



each green tick = a discriminative minimizer

largest discriminative-free (tie) zone = 196 bp

DAZ1 vs DAZ3

only 32/486 minimizers discriminative (6%) — 2.9 kb identical core where reads tie



largest discriminative-free (tie) zone = 2891 bp

Reference-only prediction matches observed read behaviour				
	discriminative minimizers	largest tie zone	PREDICTED tie-prone	OBSERVED tied reads
RABL2	58%	196 bp	0%	0 / 200 = 0%
DAZ	7%	2891 bp	~19-31%	41 / 220 = 19%

Predictor = largest discriminative-free minimizer gap vs read length. gap < read → identifiable (RABL2). gap > read → a tie zone exists (DAZ). Identity % is NOT the predictor.

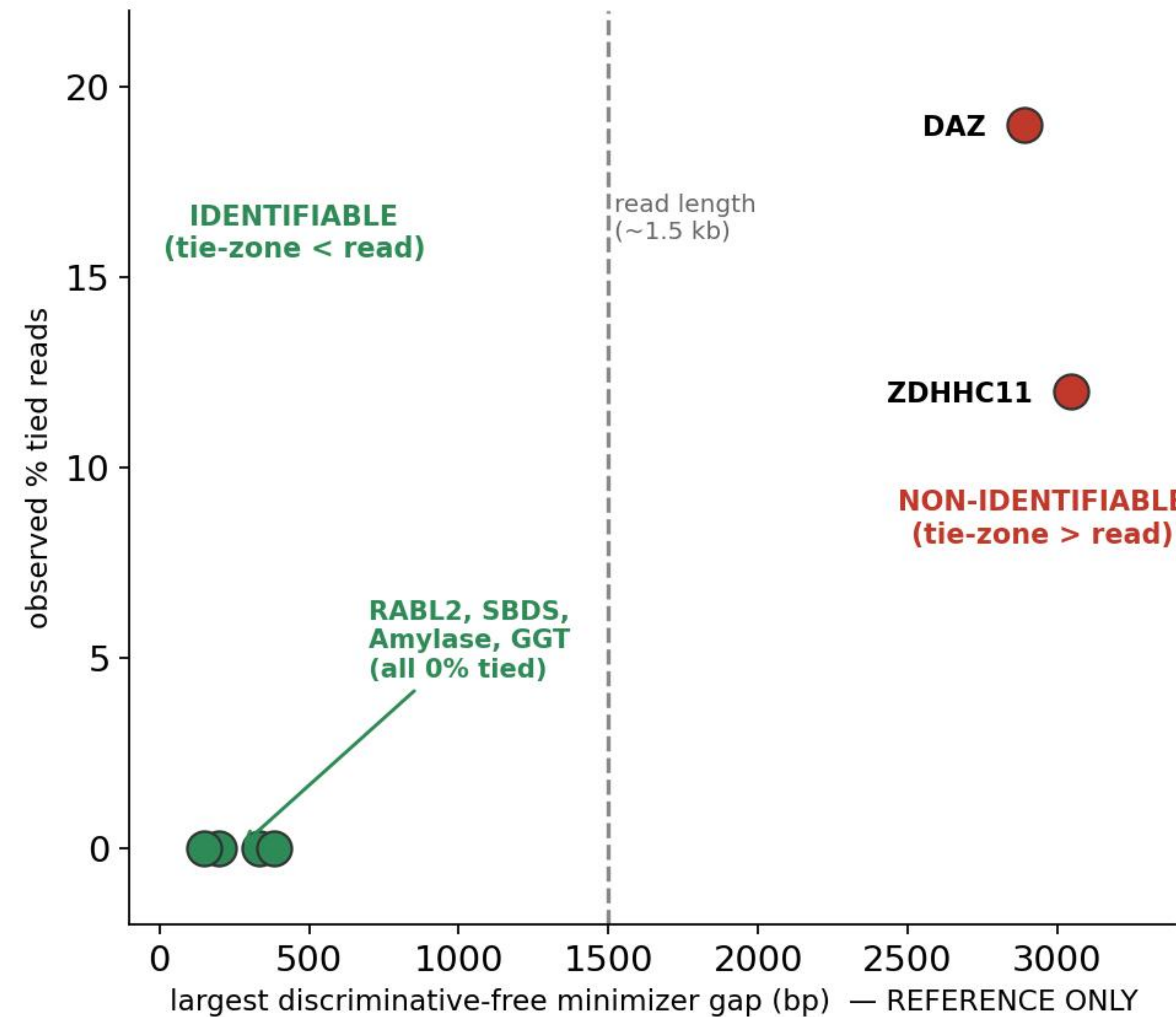
The minimizer predictor generalises — 6 families, reference-only vs reads

Prediction uses copy SEQUENCES only (no reads). Observed = % of multimapping reads that tie (equal alignment score on >1 copy).

family	copies	tie-zone	predict	obs.tied
RABL2	2	196 bp	ident ✓	0%
SBDS	2	148 bp	ident ✓	0%
Amylase	3	333 bp	ident ✓	0%
GGT	3	384 bp	ident ✓	0%
DAZ	2	2891 bp	non ✗	19%
ZDHHC11	3	3046 bp	non ✗	12%

+ rustle couldn't split

6 / 6 predictions match observed read behaviour



Identity %: RABL2 99.9% (identifiable) > DAZ (non) — identity does NOT predict. The discriminative-minimizer gap does. ZDHHC11 retrodicts the failed rustle run.